

GROUP 22

PROJECT TITLE- NGO GRANT PROPOSAL MULTIAGENT RAG BASED SYSTEM

WEEK 2 PROGRESS

Multi-Template Proposal Drafting Engine

- Replaced mock/fallback generation with an asynchronous LLM generation pipeline across three institutional grant frameworks:
 - **Standard Grant Template:** General philanthropic foundations, bilateral programs, and international grantmakers.
 - **CSR Grant Template:** Specifically aligned with corporate social responsibility mandates under Schedule VII of the Indian Companies Act, 2013.
 - **Government Grant-in-Aid (GIA) Template:** Formal administrative structure with NITI Aayog NGO Darpan compliance and administrative expenditure limits.
- Implemented section-level modular drafting (Executive Summary, Problem Statement, Objectives, Methodology & Implementation Schedule, LogFrame / M&E Framework, Budget Breakdown, Sustainability Plan, and Organizational Capability).
- Engineered dynamic prompt synthesis incorporating NGO Document Vault chunks, funder eligibility guidelines, and statutory profile metadata.

Budget Intelligence and Double-Counting Safeguards

- Developed an algorithmic budget integrity validator that cross-examines itemized expenditure sums against maximum grant caps and project totals.
- Implemented a subset-sum overlap heuristic (`_detect_subset_sum_overlap`) to automatically detect and flag duplicate cost allocations or overlapping budget line items across operational and administrative heads.
- Enforced statutory administrative expense threshold compliance (e.g., maximum 5% overhead under FCRA/CSR guidelines).

FactScore-Style Atomic Claim Verification Engine

- Engineered a claim extraction engine that decomposes drafted proposal text into discrete, atomic factual statements across entity credentials, track record, financial turnover, and beneficiary outreach metrics.
- Implemented a dual-layer hybrid verification pipeline:
 - **Deterministic Verification Layer:** Zero-hallucination validation of statutory numbers and temporal claims (NGO Darpan regex, Section 12A/80G status, FCRA certificate validity, and arithmetic operational vintage: `2026 - registration_year`).
 - **NLI Entailment Layer:** Natural Language Inference grading claims against retrieved NGO Document Vault text chunks into SUPPORTED, UNSUPPORTED, or CONTRADICTED states with direct source citations and confidence scores.
- Formulated and benchmarked the Fabrication Rate metric: **Fabrication Rate = (Unsupported Claims) / (Total Claims)**. Benchmarked on live NGO profiles: CRY produced a 41.7% fabrication rate with 5/12 unsupported claims prior to grounding; EOTO scored 50.0%.

Multi-Provider Resilient LLM Architecture

- Engineered a multi-tier fault-tolerant LLM infrastructure spanning Google Gemini (LangChain) and Groq Cloud (Llama 3.3-70B Versatile).
- Implemented intelligent model rotation across Google Gemini 3.x Flash series (gemini-3.5-flash, gemini-3.5-flash-lite, gemini-3.6-flash, gemini-3.8-flash, gemini-3.7-flash).

- Configured automated fallback to Groq Llama-3.3-70B upon quota exhaustion or API failure, providing an auxiliary 1,000 requests/day buffer.
- Implemented dynamic in-memory rate-limit and error cooldown tracking: 504 Gateway Timeout (5 min cooldown), 503 High Demand (5 min cooldown), and 429 Quota Exhaustion (2 hour cooldown).

Document Export and Institutional Publishing Engine

- Built a dual-mode document export service generating institutional-grade PDF and DOCX documents with metadata tables, formal headers, and signature blocks via ReportLab.
- Engineered a print-ready in-browser A4 CSS stylesheet with DOM visibility isolation (`body * { visibility: hidden } #printable-proposal-document { visibility: visible }`) and break-inside: avoid pagination rules preventing orphaned headers or split signature blocks.
- Developed an interactive proposal workspace supporting section-by-section live editing, markdown preview, and real-time FActScore claim audit inspection.

Testing and Verification

- Expanded the automated test suite from 16 to **24 comprehensive unit tests** covering LangGraph compilation, multi-template drafting, section-targeted revisions, statutory eligibility gates, and version auto-incrementation.
- Achieved a 100% test pass rate executed in ~1.6s with isolated offline mocking and deterministic state transition checks.
- Verified production frontend build integrity with zero compilation errors (`npm run build`).

Implementation Challenges & Plan Changes

1) LLM Quota Exhaustion & Latency Spikes (HTTP 429 & 504 Gateway Timeouts)

Challenge: Simultaneous multi-section synthesis on free-tier Gemini endpoints caused frequent 429 rate limits and 504 gateway timeouts.

Solution: Engineered a resilient multi-tier LLM architecture with Gemini 3.x model rotation and automated failover to Groq Llama 3.3-70B, combined with exponential backoff and error-specific in-memory cooldown timers.

2) Hallucination of Organizational Track Record & Temporal Metrics

Challenge: Generative models frequently hallucinated operational vintage, beneficiary counts, and financial turnover not grounded in the NGO's actual documents.

Solution: Implemented an automated atomic claim extraction pipeline coupled with a hybrid verifier: deterministic arithmetic validation for dates/vintage/registration, and strict document-grounded NLI entailment scoring each claim against the NGO Document Vault with clear citation highlights.

3) CSS Media Print Artifacts and Orphaned Document Layouts

Challenge: Standard browser printing leaked application UI elements (navigation bars, buttons, editor panels) and produced awkward page breaks cutting tables and signature blocks across pages.

Solution: Implemented DOM-level CSS visibility isolation targeting only the document preview container, combined with CSS Paged Media rules (`@page { size: A4 portrait; margin: 20mm; }`) and break-inside: avoid on all institutional blocks.